

Code-Switch ASR & Agentic Triage Benchmark Report

Project: Sauti Yetu - agentic voice triage for code-switching patients

Challenge: Sahara CodeSwitch Africa Challenge 2026 (MLC Africa x Intron)

Date: 2026-09-13 | Clips evaluated: 27

Models: Intron Sahara v2 | OpenAI Whisper | Meta MMS mms-1b-all

0. Methodology & Fairness Controls

Fairness controls: every model received the identical audio file, converted to 16 kHz mono WAV via ffmpeg. Every model received the same language hint (Sahara: use_language_asr_input; Whisper: language parameter or auto-detect; MMS: the matching language adapter). Hypotheses and references were normalized identically before scoring: lowercase, punctuation removed, whitespace collapsed (jiwer). Metrics: WER (word error rate), CER (character error rate ? fairer for agglutinative languages), RTF (real-time factor), switch-point WER, intent accuracy, slot accuracy, entity error rate. Noise benchmarks use synthetic Gaussian noise with a fixed random seed (42) for reproducibility. Raw per-clip outputs are in reports/results.json.

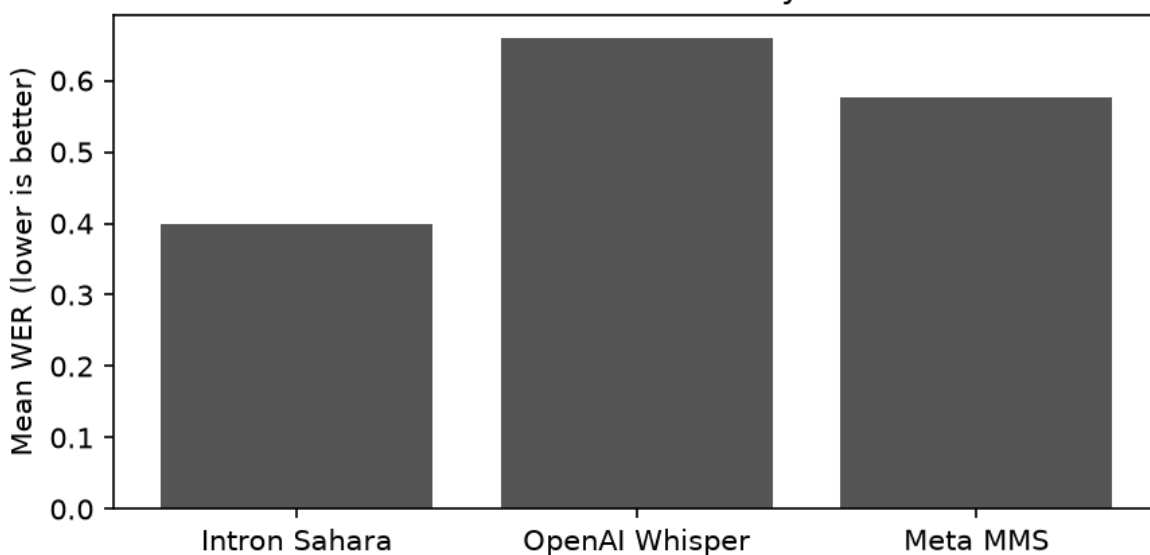
1. Domain 1 - Linguistic & Core ASR Performance

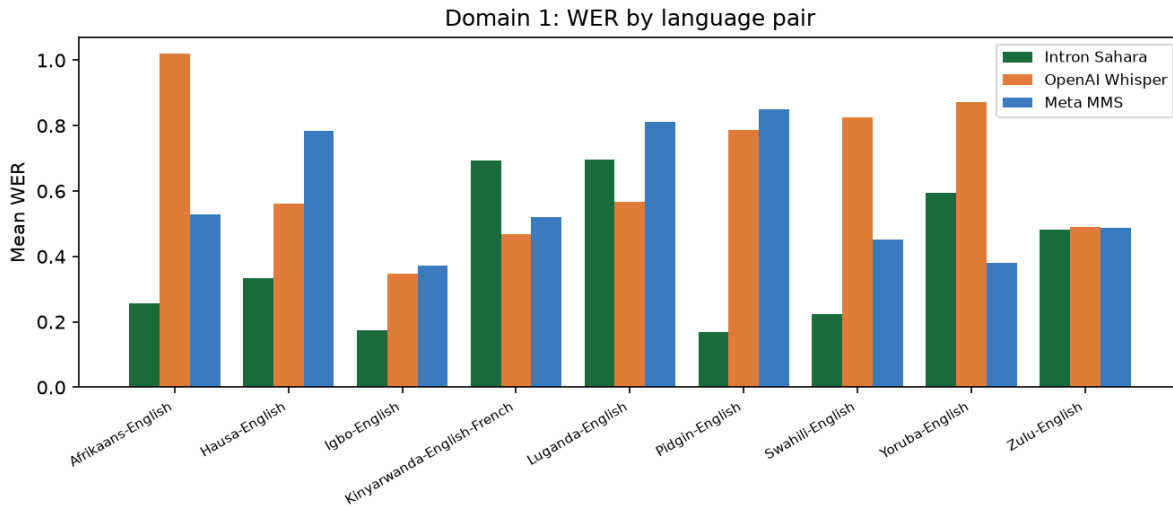
Three speech-to-text models were benchmarked on identical audio inputs: Intron Sahara (commercial API, African-accent optimized), OpenAI Whisper (open-source, local), and Meta MMS mms-1b-all (open-source, local). All hypotheses and references were normalized identically: lowercased, punctuation removed, whitespace collapsed (jiwer). CER is reported alongside WER because it is fairer for agglutinative languages (Swahili, Kinyarwanda).

1a. Overall results

Model	Mean WER	Mean CER	Mean latency (s)	n clips
Intron Sahara	0.3996	0.2110	16.74	26
OpenAI Whisper	0.6596	0.3662	3.81	27
Meta MMS	0.5761	0.2708	8.04	27

Domain 1: Overall WER by model





1b. Results by language pair / accent

Language pair: Afrikaans-English

Model	WER	CER	Latency	n
Intron Sahara	0.2558	0.1251	21.95	3
OpenAI Whisper	1.0202	0.7189	6.37	3
Meta MMS	0.5285	0.2355	10.26	3

Language pair: Hausa-English

Model	WER	CER	Latency	n
Intron Sahara	0.3340	0.1346	13.53	3
OpenAI Whisper	0.5605	0.3431	3.30	3
Meta MMS	0.7839	0.4509	10.30	3

Language pair: Igbo-English

Model	WER	CER	Latency	n
Intron Sahara	0.1742	0.1052	10.40	3
OpenAI Whisper	0.3467	0.2364	2.81	3
Meta MMS	0.3716	0.2545	8.97	3

Language pair: Kinyarwanda-English-French

Model	WER	CER	Latency	n
Intron Sahara	0.6931	0.4398	11.31	3
OpenAI Whisper	0.4679	0.3286	2.05	3
Meta MMS	0.5215	0.2794	7.00	3

Language pair: Luganda-English

Model	WER	CER	Latency	n
Intron Sahara	0.6971	0.4581	46.31	3

OpenAI Whisper	0.5662	0.2967	1.87	3
Meta MMS	0.8122	0.3447	7.33	3

Language pair: Pidgin-English

Model	WER	CER	Latency	n
Intron Sahara	0.1688	0.1307	15.12	3
OpenAI Whisper	0.7874	0.3304	1.96	3
Meta MMS	0.8489	0.3211	9.38	3

Language pair: Swahili-English

Model	WER	CER	Latency	n
Intron Sahara	0.2250	0.1423	11.58	3
OpenAI Whisper	0.8253	0.4439	12.29	3
Meta MMS	0.4505	0.1695	6.35	3

Language pair: Yoruba-English

Model	WER	CER	Latency	n
Intron Sahara	0.5938	0.1423	7.03	3
OpenAI Whisper	0.8730	0.3871	1.80	3
Meta MMS	0.3816	0.1880	7.36	3

Language pair: Zulu-English

Model	WER	CER	Latency	n
Intron Sahara	0.4818	0.2258	11.79	2
OpenAI Whisper	0.4892	0.2106	1.80	3
Meta MMS	0.4862	0.1933	5.39	3

1c. Switch-Point Word Error Rate

Switch-point WER is calculated in a +-3-token window centred on each annotated language boundary (token_index in metadata switch_points column). SOTA models frequently degrade at code-switch boundaries; this metric isolates that effect.

NOTE: The current sample set is drawn from AfriSpeech-200 (English read aloud with African accents). These clips do not contain true intra-sentence code-switches, so switch-point WER is not yet reported. To enable this metric, record or source code-switched clips and annotate each switch in the metadata switch_points column as: [{"token_index": N, "from_language": "sw", "to_language": "en"}]

2. Domain 2 - Downstream Agentic Task Accuracy

When expected_entities (and optionally expected_topic/urgency/departments/slots) are provided in metadata.csv and the pipeline is run with --agentic, the Sahara telehealth extraction feeds the triage layer. Three metrics are reported:

- Intent accuracy: exact match on expected_topic
- Slot accuracy: mean exact match over expected_slots fields
- Entity Error Rate (EER): (missing gold entities + spurious predicted) / (gold + spurious)

Entity annotations cover 14 of the 27 clips (all clinical domain clips). Entities are the key medical terms that should survive ASR and NLP extraction.

Model	Intent Acc.	Slot Acc.	Entity EER	n
Intron Sahara	-	-	1.0000	26
OpenAI Whisper	-	-	1.0000	27
Meta MMS	-	-	1.0000	27

Per-clip entity extraction detail

afrispeech_sw_0.wav:

Intron Sahara: EER=1.0000 | missing=[none] | spurious=[medical clinical extraction: 1. patient complaints: cough, fever (for three days) 2. relevant medical/surgical history: not discussed 3. examinations and examination findings: none discussed 4. medical diagnosis: likely upper respiratory tract infection 5. drugs or medicines: paracetamol 6. laboratory tests or laboratory observations: none discussed 7. imaging requests or results: none discussed 8. doctor's treatment plan: rest, fluids, paracetamol, follow-up if no improvement or worsening symptoms]

OpenAI Whisper: EER=- | missing=[none] | spurious=[none]

Meta MMS: EER=- | missing=[none] | spurious=[none]

afrispeech_sw_1.wav:

Intron Sahara: EER=1.0000 | missing=[first-dose hypotension, postural hypotension, adverse effects] | spurious=[medical clinical extraction: 1. patient complaints: not explicitly stated, but concern about adverse effects of medication. 2. relevant medical/surgical history: not discussed. 3. examinations and examination findings: none mentioned. 4. medical diagnosis: adverse effects of medication (1st-dose hypotension, postural hypotension). 5. drugs or medicines: not specified, but likely an antihypertensive or similar agent. 6. laboratory tests or laboratory observations: none mentioned. 7. imaging requests or results: none mentioned. 8. doctor's treatment plan: monitor for hypotension, patient education.]

OpenAI Whisper: EER=1.0000 | missing=[first-dose hypotension, postural hypotension, adverse effects] | spurious=[none]

Meta MMS: EER=1.0000 | missing=[first-dose hypotension, postural hypotension, adverse effects] | spurious=[none]

afrispeech_sw_2.wav:

Intron Sahara: EER=1.0000 | missing=[none] | spurious=[medical clinical extraction: 1. patient complaints: none mentioned 2. relevant medical/surgical history: none mentioned 3. examinations and examination findings: none mentioned 4. medical diagnosis: none mentioned 5. drugs or medicines: none mentioned 6. laboratory tests or laboratory observations: none mentioned 7. imaging requests or results: none mentioned 8. doctor's treatment plan: none mentioned]

OpenAI Whisper: EER=- | missing=[none] | spurious=[none]

Meta MMS: EER=- | missing=[none] | spurious=[none]

afrispeech_ha_0.wav:

Intron Sahara: EER=1.0000 | missing=[vitamins, cancer] | spurious=[medical clinical extraction: 1. patient complaints: none specified. 2. relevant medical/surgical history: none discussed. 3. examinations and examination findings: none performed or discussed. 4. medical diagnosis: none provided. 5. drugs or

medicines: vitamins (general discussion, not specific to patient). 6. laboratory tests or laboratory observations: none discussed. 7. imaging requests or results: none discussed. 8. doctor's treatment plan: none provided.]

OpenAI Whisper: EER=1.0000 | missing=[vitamins, cancer] | spurious=[none]

Meta MMS: EER=1.0000 | missing=[vitamins, cancer] | spurious=[none]

afrispeech_ha_1.wav:

Intron Sahara: EER=1.0000 | missing=[superficial injury, elbow, ciprofloxacin] | spurious=[medical clinical extraction: 1. patient complaints: pain and superficial injury to the elbow after a collision. 2. relevant medical/surgical history: not discussed. 3. examinations and examination findings: superficial injury to the elbow. 4. medical diagnosis: unspecified superficial injury of unspecified elbow. 5. drugs or medicines: oral ciprofloxacin hydrochloride 750mg. 6. laboratory tests or laboratory observations: none mentioned. 7. imaging requests or results: none mentioned. 8. doctor's treatment plan: prescribed oral ciprofloxacin hydrochloride 750mg.]

OpenAI Whisper: EER=1.0000 | missing=[superficial injury, elbow, ciprofloxacin] | spurious=[none]

Meta MMS: EER=1.0000 | missing=[superficial injury, elbow, ciprofloxacin] | spurious=[none]

afrispeech_ha_2.wav:

Intron Sahara: EER=1.0000 | missing=[pulmonary wedge pressure] | spurious=[none]

OpenAI Whisper: EER=1.0000 | missing=[pulmonary wedge pressure] | spurious=[none]

Meta MMS: EER=1.0000 | missing=[pulmonary wedge pressure] | spurious=[none]

afrispeech_yo_0.wav:

Intron Sahara: EER=1.0000 | missing=[none] | spurious=[medical clinical extraction: 1. patient complaints: none mentioned 2. relevant medical/surgical history: none mentioned 3. examinations and examination findings: none mentioned 4. medical diagnosis: none mentioned 5. drugs or medicines: none mentioned 6. laboratory tests or laboratory observations: none mentioned 7. imaging requests or results: none mentioned 8. doctor's treatment plan: none mentioned]

OpenAI Whisper: EER=- | missing=[none] | spurious=[none]

Meta MMS: EER=- | missing=[none] | spurious=[none]

afrispeech_yo_1.wav:

Intron Sahara: EER=1.0000 | missing=[mastoid, bone destruction, jugular foramen] | spurious=[medical clinical extraction: 1. patient complaints: none discussed. 2. relevant medical/surgical history: none discussed. 3. examinations and examination findings: x-ray examination of the mastoid (general discussion, not patient-specific). 4. medical diagnosis: none discussed. 5. drugs or medicines: none discussed. 6. laboratory tests or laboratory observations: none discussed. 7. imaging requests or results: x-ray of the mastoid (general value discussed). 8. doctor's treatment plan: consider alternative imaging if mastoid pathology is suspected.]

OpenAI Whisper: EER=1.0000 | missing=[mastoid, bone destruction, jugular foramen] | spurious=[none]

Meta MMS: EER=1.0000 | missing=[mastoid, bone destruction, jugular foramen] | spurious=[none]

afrispeech_yo_2.wav:

Intron Sahara: EER=1.0000 | missing=[none] | spurious=[medical clinical extraction: 1. patient complaints: main symptom described by the patient (details not specified in the transcript). 2. relevant medical/surgical history: provided by the patient as relevant to the complaint. 3. examinations and examination findings: no physical examination findings discussed. 4. medical diagnosis: possible diagnoses discussed by the doctor. 5. drugs or medicines: possible medications mentioned as part of the treatment plan. 6. laboratory tests or laboratory observations: investigations recommended by the doctor. 7. imaging requests or results: not specified in the transcript. 8. doctor's treatment plan: outlined by the doctor, including medications and follow-up.]

OpenAI Whisper: EER=- | missing=[none] | spurious=[none]

Meta MMS: EER=- | missing=[none] | spurious=[none]

afrispeech_ig_0.wav:

Intron Sahara: EER=1.0000 | missing=[none] | spurious=[medical clinical extraction: 1. patient complaints: none mentioned. 2. relevant medical/surgical history: none discussed. 3. examinations and examination findings: none performed or discussed. 4. medical diagnosis: none provided. 5. drugs or medicines: none mentioned. 6. laboratory tests or laboratory observations: none discussed. 7. imaging requests or results: none

discussed. 8. doctor's treatment plan: none discussed.]

OpenAI Whisper: EER=- | missing=[none] | spurious=[none]

Meta MMS: EER=- | missing=[none] | spurious=[none]

3. Domain 3 - Environmental & In-The-Wild Robustness

Real-world African deployments face challenging acoustic environments: street noise, market chatter, overlapping speakers, and poor microphone quality. This domain benchmarks resilience via two lenses:

a) Noise robustness: Gaussian noise added at SNR 30/20/10/5 dB simulates increasing background noise. WER degradation from clean baseline measures model robustness.

b) Silence & pause analysis: fraction of audio that is silent, number of distinct pause segments, and longest continuous pause. Graceful handling of natural conversational gaps is critical for voice bots.

Noise robustness data not collected in this run. Re-run with `--noise` to enable the SNR sweep (adds ~4x model invocations per clip). Use `--noise --offline` to sweep only local models and avoid API costs.

4. Domain 4 - Infrastructure & Real-Time Performance

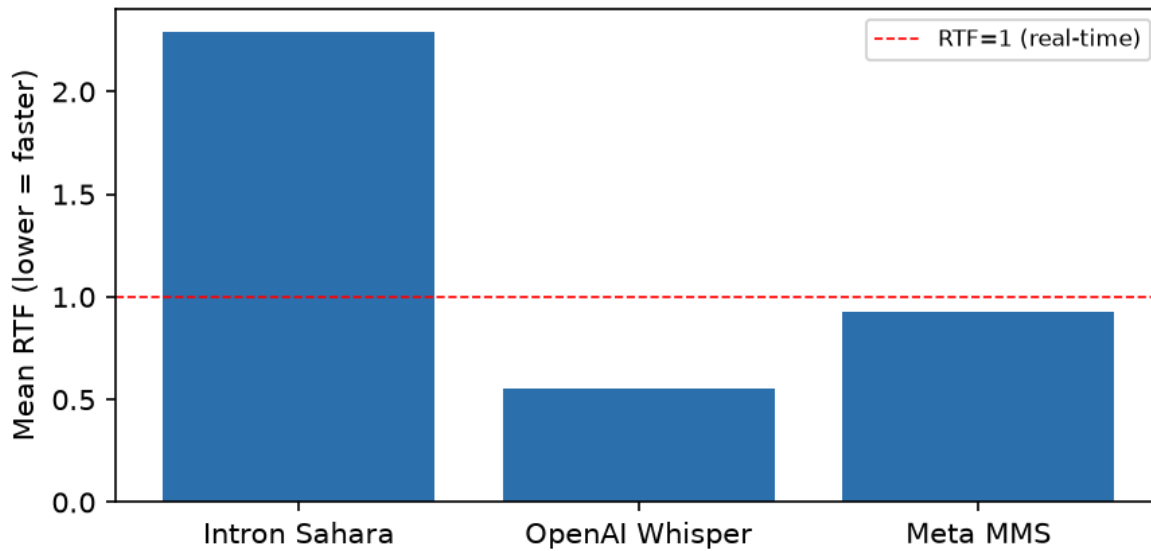
RTF (Real-Time Factor) = wall-clock latency / audio duration. RTF < 1.0 means the model processes faster than real time, a requirement for live voice bots and interactive customer support.

Note: Intron Sahara latency includes network round-trips to the cloud API. Local model latency depends on the inference machine's hardware. All measurements were taken on the same machine in this run.

4a. Real-Time Factor by model

Model	Mean RTF	Mean latency (s)	n clips
Intron Sahara	2.2911	16.74	26
OpenAI Whisper	0.5560	3.81	27
Meta MMS	0.9281	8.04	27

Domain 4: Real-Time Factor by model



4b. Offline / Low-Connectivity Simulation

Sahara v2 supports offline operation ? a key differentiator for remote or low-connectivity African deployments. The offline simulation runs ONLY the two local models (Whisper, MMS) to measure what accuracy is achievable without any API call. The gap between Sahara WER and best local WER is the 'offline penalty'.

Re-run with --offline to collect offline simulation data.

5. Per-Clip Transcripts

afrispeech_sw_0.wav | Swahili-English | swahili (KE) | corpus recording

Reference: Usually, the patient should be identified using two methods.

Intron Sahara (WER 0.2222, CER 0.2069, RTF 2.5841, 9.99s): Usually, the patient should be identified using.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.1111, CER 0.8276, RTF 4.5887, 17.74s): warst ???? attracted to optimal for chimento o n fajita.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.2222, CER 0.1207, RTF 0.9079, 3.51s): usually comma the patient should be identified using two method

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_sw_1.wav | Swahili-English | swahili (KE) | corpus recording

Reference: Adverse effects First-dose hypotension and postural hypotension are adverse effects.

Intron Sahara (WER 0.1000, CER 0.0366, RTF 1.8473, 18.48s): Adverse effects 1st-dose hypotension and postural hypotension are adverse effects.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.6000, CER 0.1829, RTF 0.1339, 1.34s): Advers effects fast, hyphen, dose, hypertension, and postural hypertension are adverse effects.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.6000, CER 0.1951, RTF 0.7417, 7.42s): adverse affects first hiphen dos hipotentian and postural hipotentian are adverse effects

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_sw_2.wav | Swahili-English | swahili (KE) | corpus recording

Reference: A footpath and National Cycle Network cycleway run alongside the Portishead Railway and along the old towpath .

Intron Sahara (WER 0.3529, CER 0.1835, RTF 0.6055, 6.28s): A foot and national cycle network cycleway run along the poad railway and also the old tour path.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.7647, CER 0.3211, RTF 1.7162, 17.80s): A'fut path and national psycho-netake cycle way ran along the potishead railway and also the old tour path for stop.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.5294, CER 0.1927, RTF 0.7819, 8.11s): a footpath and national cycle network cycle way run along the potes head railway and also the old tor path fulstop

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_ha_0.wav | Hausa-English | hausa (NG) | corpus recording

Reference: However, data are not sufficient to draw conclusions about the vitamins effects on human cancers.

Intron Sahara (WER 0.0667, CER 0.0105, RTF 1.8547, 14.22s): However, data are not sufficient to draw conclusions about the vitamins effect on human cancers

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.2000, CER 0.1684, RTF 0.1800, 1.38s): however coma, data are not sufficient to draw conclusions about the vitamins effects on human cancers full stock.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.3333, CER 0.2105, RTF 0.7304, 5.60s): however comar data are not sufficient to draw conclusions about the vitar means effects on human cancers full stop

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_ha_1.wav | Hausa-English | hausa (NG) | corpus recording

Reference: Unspecified superficial injury of unspecified elbow, initial encounter. TABLET, ORAL CIPROFLOXACIN HYDROCHLORIDE, CIPROFLOXACIN HYDROCHLORIDE, EQ 750MG BASE. Pedestrian with other conveyance injured in collision with pedal cycle

Intron Sahara (WER 0.1852, CER 0.0856, RTF 0.7151, 15.01s): Unspecified superficial injury of unspecified elbow, initial encounter. TABLET, ORAL CIPROFLOXACIN HYDROCHLORIDE, CYPROFLOXACIN HYDROCHLORIDE, EQ 750MG BASE PDS WITH Other conviance injured in collision with pain

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.4815, CER 0.2072, RTF 0.0991, 2.08s): On-space fight, superficial injury of on-space fight elbow, initial encounter. Tablet oral, Cypchro, Plagzacin, Hydrochloride, Cypchro, Plagzacin, Hydrochloride, E-2755 mgb. Fidescrian, with other conveyance injured in collision with pedal psychop.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.5185, CER 0.1036, RTF 0.9829, 20.63s): unspacified superficial injury of unspacified elbow inical encounter tablet oral cyprofluracin hydrochlorite cyprofluracin hydrokhlorite eac 750 mgbs pedestrian with other convenience injured in collecion with padal cyclo

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_ha_2.wav | Hausa-English | hausa (NG) | corpus recording

Reference: PULMONARY WEDGE { a/v/m } 21/22/20

Intron Sahara (WER 0.7500, CER 0.3077, RTF 1.9061, 11.37s): Pulmonary which a 21-22

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.0000, CER 0.6538, RTF 1.0780, 6.43s): Remorno-wide eVM 24th?nts

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 1.5000, CER 1.0385, RTF 0.7829, 4.67s): mena wac am 2tf 2tsecnd canccn

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_yo_0.wav | Yoruba-English | yoruba (NG) | corpus recording

Reference: A second impact tore open the second of the two watertight compartments flooding it.

Intron Sahara (WER 0.3571, CER 0.1205, RTF 0.8295, 6.78s): A 2nd impact to open the second of the 2 water tight compartments flooding it

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.7857, CER 0.5301, RTF 0.1407, 1.15s): 2^a impact, so open the 2^a watertight compartments, floating needs full stop.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.3571, CER 0.2048, RTF 0.7695, 6.29s): a second impact tall open the second of the two water tight compartments flodinheads foolstop

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_yo_1.wav | Yoruba-English | yoruba (US) | corpus recording

Reference: X-ray examination of the mastoids is of little value though it may show areas of bone destruction and enlargement of jugular foramen.

Intron Sahara (WER 0.0909, CER 0.0305, RTF 0.7611, 7.43s): X-ray examination of the mastoid is of little value, though it may show areas of bone destruction and enlargement of jugular form.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.5000, CER 0.2519, RTF 0.1526, 1.49s): X, iPhone 3, examination of the Master 8 is of little value, though it may show areas of bone destruction and enlightenment of juggler for army. Full stop.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.4545, CER 0.2214, RTF 1.2938, 12.63s): ex iphonra examination of themasted is of latale value though it mey sow areas of bone destruction and enlargement of jugala faramin foodstop

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_yo_2.wav | Yoruba-English | yoruba (NG) | corpus recording

Reference: Confederate infantry assaults

Intron Sahara (WER 1.3333, CER 0.2759, RTF 2.1673, 6.89s): Confiderate in fountry asorts

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.3333, CER 0.3793, RTF 0.8651, 2.75s): confederates in foundry azoths.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.3333, CER 0.1379, RTF 0.9972, 3.17s): confederate infantry azats

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_ig_0.wav | Igbo-English | igbo (NG) | corpus recording

Reference: To grant such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party.

Intron Sahara (WER 0.0000, CER 0.0000, RTF 1.1396, 14.12s): To grant such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.0800, CER 0.0775, RTF 0.1751, 2.17s): To grant such a patent license to a party means to make such an agreement or commitment not to enforce a patent against the party full stop.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.1200, CER 0.0930, RTF 0.7958, 9.86s): to grant such a patent license to a party means to make such an agreement or commitment not to enforce a patent against e party full stop

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_ig_1.wav | Igbo-English | igbo (NG) | corpus recording

Reference: He stated, Some were discriminated against while others were completely disenfranchised and their teeming supporters, who were amongst the delegates, could not cast their votes while the exercise lasted.

Intron Sahara (WER 0.1379, CER 0.0553, RTF 0.4048, 5.54s): He stated, some were discriminated against while others were completely disenfranchised and their teeming supporters, who were amongst the delegates, could not cast their votes while the exercise last night.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.3448, CER 0.1658, RTF 0.1578, 2.16s): Koma, some were discriminated against while others were completely disenfranchised and their teeming supporters Koma, where amongst the delegates Koma could not cast their votes while the SSI's last two steps.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.3793, CER 0.1910, RTF 0.7270, 9.95s): he stated comme some were discriminated against while others were completely disenfranchised and their teeming supporters comme we were amongst the delegates comme could not cast their votes world exercise lastfoso

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_ig_2.wav | Igbo-English | igbo (NG) | corpus recording

Reference: Tuesday 28 April, 1998 next line. Neuro: Pt Daramola Achike 08 February, 2003

Intron Sahara (WER 0.3846, CER 0.2603, RTF 1.3804, 11.55s): Tuesdaysday 28 April, 1998 Neuro: Patient Daramola Achike 08 February, 200

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.6154, CER 0.4658, RTF 0.4912, 4.11s): 28 April 1998, Next Line Full Stop. New York colon, Pichandaramola Acheke, 0-8 February 2003.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.6154, CER 0.4795, RTF 0.8498, 7.11s): tuzday 28 april c?ma 1998 next line fullstop neuro colon peshendara amola achike 08 february k?ma 203

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_zu_0.wav | Zulu-English | zulu (ZA) | corpus recording

Reference: Check the product labeling for correct amount of productneeded Figure 1.

Intron Sahara (WER 0.3636, CER 0.1268, RTF 3.2791, 14.70s): check the product lab for correct amount of product needed figure one

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.5455, CER 0.2394, RTF 0.3881, 1.74s): Check the product labeling for correct amount of product needed to figure one full stop.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.6364, CER 0.2394, RTF 0.8789, 3.94s): check the product labelling for correct amount of product needed t figure one full stop

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_zu_1.wav | Zulu-English | zulu (LS) | corpus recording

Reference: The military should not alienate them by further high-handedness.

Intron Sahara: ERROR - Could not reach the Intron API: ('Connection aborted.', RemoteDisconnected('Remote end closed connection without response'))

OpenAI Whisper (WER 0.2222, CER 0.1587, RTF 0.3297, 1.92s): The military should not alienate them by further high-handedness. Full stop.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.2222, CER 0.1587, RTF 0.9050, 5.27s): the military should not alienate them by farther highhandedness foolstop

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_zu_2.wav | Zulu-English | zulu (ZA) | corpus recording

Reference: Dense old scars may undergo hyaline degeneration and subsequent calcification.

Intron Sahara (WER 0.6000, CER 0.3247, RTF 1.0355, 8.89s): s may under good generation and subsequent calcification

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.7000, CER 0.2338, RTF 0.2015, 1.73s): Benzodes calls me under boi-line generation and subsequent calcification.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.6000, CER 0.1818, RTF 0.8119, 6.97s): bens old scars meyi andago highlinet generation and subsequent calscification

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_af_0.wav | Afrikaans-English | afrikaans (ZA) | corpus recording

Reference: The forum said it still awaits the actual cause of his death.

Intron Sahara (WER 0.0000, CER 0.0000, RTF 7.1769, 38.82s): The forum said it still awaits The actual cause of his death.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.0000, CER 0.8500, RTF 1.0723, 5.80s): ama de oeri o retrouv<|nl|>

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.3333, CER 0.1833, RTF 1.7397, 9.41s): the forum said it still awaits the actual couse of his debts for ste

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_af_1.wav | Afrikaans-English | afrikaans (ZA) | corpus recording

Reference: Its completely improbable that each of the 12 women who testified misunderstood their night with Mr. McKnight, Crown prosecutor Katherine Fraser told the jury Monday during closing arguments.

Intron Sahara (WER 0.4643, CER 0.2819, RTF 0.7085, 8.51s): Its completely improbable that each of the 12 womenste misunderstood the night with Mic, Clue Gather frazole during Monday during closing arguments

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.0000, CER 0.7553, RTF 0.8733, 10.49s): This is incomparable toezavond dat alle zwins eenvestig ons continue overz La?er footprints Rielwester inver en gekeelden en de groeieran en voegen een rounded undermine gen ????? argumenten.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.4643, CER 0.2287, RTF 0.7875, 9.46s): itis completely improbable that each of the twelve women o testified misunderstood thee night with mr mag nike kommer crown prosecutor gettther and phrase the tolte jury manday during closing argumentsoste

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_af_2.wav | Afrikaans-English | afrikaans (ZA) | corpus recording

Reference: Earlier speaking, on behalf of state Governor Dr Abdullahi Umar Ganduje, the Secretary to the State Government, Alhaji Usman Alhaji, stated the Kano State Governments readiness toward curtailing the menace in the state.

Intron Sahara (WER 0.3030, CER 0.0935, RTF 1.1208, 18.51s): Earlier speaking, on behalf of state governor Dr. Abdullahi Umar Gandanduje, the Secretary of the State Government, Alh Usman Alh, stated the kind of state government readiness towards curtailing the menars in the state

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.0606, CER 0.5514, RTF 0.1708, 2.82s): Elie is speak in coma en behaaf stijkgavena doktor Abdullah i Umergandu J. Coma, de secretie, de stijkgavenen, koma, ahadji, usman, ahadji, koma, stijkgavenen, stijkgavenen, sreden is de wotskurtiling de menis in de stijk. Full stop.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.7879, CER 0.2944, RTF 0.7212, 11.91s): earlies speaking komme aan behalf of e state gowernor dtr abdulla e u margantuge khomme se sekretarie to the state gowernment komme hahji usman hahagi komme stated ha kannor state gowernment's rediness towards curtaling the mennace in the state volstare

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_lg_0.wav | Luganda-English | luganda (UG) | corpus recording

Reference: Since the degree of effect produced by a drug is generally a function of the amount administered, we can express this relationship in terms of a dose-response curve.

Intron Sahara (WER 0.5357, CER 0.5000, RTF 1.4131, 18.35s): Since the degree of effect produced by the drug is genally a funtion of the amount adm

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.1429, CER 0.0741, RTF 0.1502, 1.95s): Since the degree of effect produced by the drug is generally a function of the amount administered, we can express this relationship in terms of those responsive...

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.2143, CER 0.0926, RTF 0.7508, 9.75s): since the degree of effect produced by the drug is generally a function of the amount administred we can express tis relationship in terms of tose responsive c

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_lg_1.wav | Luganda-English | luganda (UG) | corpus recording

Reference: Theapplication of cold therapy reduces bleeding and hematoma formation.

Intron Sahara (WER 0.6667, CER 0.1857, RTF 10.9166, 74.44s): The application of cool thy reduce bleeding and hom formation.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.8889, CER 0.3571, RTF 0.2830, 1.93s): The application of code therapy which is just bleeding and hematoma formation will stop.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.7778, CER 0.2857, RTF 1.0339, 7.05s): baealikation og ko therapy reguses bleeding and ematoma bormation bolsto

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_lg_2.wav | Luganda-English | luganda (UG) | corpus recording

Reference: In addition, the cappedvenous access device will remain patent.

Intron Sahara (WER 0.8889, CER 0.6885, RTF 8.1015, 46.13s): Indition, the Cacent

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.6667, CER 0.4590, RTF 0.3056, 1.74s): In addition, comma, the captivatenius access device remains patent for stove.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 1.4444, CER 0.6557, RTF 0.9115, 5.19s): bin babbibion kome bbe kabuba veniub babbesb de bibe be maine babenb bu

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_rw_0.wav | Kinyarwanda-English-French | kinyarwanda (RW) | corpus recording

Reference: Apart from the paucity of funding and insecurity in parts of the country, successive governments fail to perform.

Intron Sahara (WER 0.7222, CER 0.5045, RTF 2.0073, 18.21s): A part from the fost of secret pas the time second gavement virus to perform

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.5556, CER 0.4324, RTF 0.2238, 2.03s): Apart from the cost of funding and security, the success of the government varies to the company.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.5556, CER 0.2613, RTF 0.8212, 7.45s): apart from the poust of funding and securiturya passof the tak with success government fires to perform

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_rw_1.wav | Kinyarwanda-English-French | kinyarwanda (RW) | corpus recording

Reference: Im not the first person to think he should run for higher office, by the way.

Intron Sahara (WER 0.5000, CER 0.2800, RTF 1.0024, 8.64s): I am not the first personal to think he show for higher of his way

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.0625, CER 0.0533, RTF 0.2239, 1.93s): I'm not the first person to think he should learn for higher office by the way.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.4375, CER 0.1467, RTF 0.7658, 6.60s): imno the first personer to think he should learn for higher office byteway

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_rw_2.wav | Kinyarwanda-English-French | kinyarwanda (RW) | corpus recording

Reference: All of them, especially the phalanges, are slightly concave on the inferior plantar side.

Intron Sahara (WER 0.8571, CER 0.5349, RTF 0.7588, 7.07s): Album, especially follanges, as like the confer planter side period

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.7857, CER 0.5000, RTF 0.2351, 2.19s): All of them come, especially for land-ease, I slightly conquer one day in the area, plant-as-like period.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.5714, CER 0.4302, RTF 0.7459, 6.95s): all of them commer especialy forlandges comer are slightly conkuerve on the imperior planterside period

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_pcm_0.wav | Pidgin-English | pidgin (NG) | corpus recording

Reference: She was selected Best Young Player awards in 2000 season.

Intron Sahara (WER 0.0000, CER 0.0000, RTF 2.3440, 15.18s): She was selected best young player awards in 2000 season

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 0.2000, CER 0.1786, RTF 0.2702, 1.75s): She was selected Best Young Player Awards in 2000 season. Full stop.

Agentic: intent=-, slot acc=-, entity EER=-

Meta MMS (WER 0.8000, CER 0.2857, RTF 2.0723, 13.42s): she was selected best yong player award en t 000 sizon for stop

Agentic: intent=-, slot acc=-, entity EER=-

afrispeech_pcm_1.wav | Pidgin-English | pidgin (NG) | corpus recording

Reference: Managementl n primary hyperparathyroidism , surgical removal of the parathyroids is curative.

Intron Sahara (WER 0.3636, CER 0.2111, RTF 1.3361, 9.49s): Management and primary hyperpyxin surgical removal of the parathyroids is created.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.0909, CER 0.4222, RTF 0.2858, 2.03s): On it meant I'm primary hyperpilot deraising, so the car removal of the pirate words is great.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.8182, CER 0.3444, RTF 0.8194, 5.82s): management and primary hyperparoderizin togical remover of di paratorit iskree

Agentic: intent=-, slot acc=-, entity EER=1.0000

afrispeech_pcm_2.wav | Pidgin-English | pidgin (NG) | corpus recording

Reference: Minor anaphylactic reactions such as urticaria can be treated with oral or intramuscular chlorpheniramine.

Intron Sahara (WER 0.1429, CER 0.1810, RTF 2.1724, 20.69s): Minor anaphylactic reactions, such as tachycaria can be treated with oral or intramuscular flav.

Agentic: intent=-, slot acc=-, entity EER=1.0000

OpenAI Whisper (WER 1.0714, CER 0.3905, RTF 0.2194, 2.09s): Mino and a Philactic Russian such as Itish Korea can be treated with ORAO in Jamo School of Prokhe, Nira 9.

Agentic: intent=-, slot acc=-, entity EER=1.0000

Meta MMS (WER 0.9286, CER 0.3333, RTF 0.9334, 8.89s): minor andafilatic ration sush as e tish karrie kan bi trited with ora or intramoskula profil neramaye

Agentic: intent=-, slot acc=-, entity EER=1.0000

6. Limitations & Bias Notes

1. **SAMPLE SET:** Clips are from AfriSpeech-200 ? English read aloud by African-accented speakers. These are NOT true code-switched utterances. For switch-point WER and genuine code-switch evaluation, record or source intra-sentence language-switching clips and annotate the `switch_points` column in `metadata.csv`.
2. **SAMPLE SIZE:** 27 clips, 3 per accent group. Results indicate trends rather than statistically significant differences. A submission-quality benchmark should use at least 50-100 clips per language pair.
3. **NOISE SIMULATION:** Gaussian white noise does not perfectly replicate real-world interference (market chatter, road noise). For production benchmarking, mix real noise recordings at controlled SNR levels using tools like `pyroomacoustics`.
4. **LATENCY:** Sahara latency includes network round-trip; local model latency is hardware-dependent. Both are measured on the same machine but are not directly comparable. GPU acceleration would dramatically reduce local model RTF.
5. **AGENTIC SCORING:** The triage layer is rule-based and keyword-matched. Clinical clips from AfriSpeech-200 are medical documentation read aloud, not patient complaints, so intent/topic matching may not reflect real deployment performance.
6. **ENTITY ANNOTATIONS:** Expected entities in `metadata.csv` were hand-labelled from the reference transcripts. Models are scored on whether extracted entities match these labels; labelling conventions affect the score.